

MACHINE LEARNING — LAB MANUAL

Lab 2: From Raw Data to ML-Ready Data — Data Cleaning & Exploratory Data Analysis

A 3-Hour Independent Practical Lab — BS Computer Science, 5th Semester

Duration: 3 Hours (independent, hands-on lab)

Prerequisites: Hands-on Python, NumPy, Pandas & Matplotlib from prior labs

Tools Required: Python 3.x, Pandas, NumPy, Matplotlib/Seaborn, Jupyter Notebook / Google Colab, a GitHub account

Includes: Task-driven investigation, data cleaning, EDA, target & feature analysis, ML-readiness checklist, and a graded rubric

Learning Outcomes

This lab is NOT a Python/Pandas tutorial — you already know the syntax. The goal is to practice the reasoning a data scientist applies to an unfamiliar, real-world dataset before any modeling begins. By the end of this lab you should be able to independently:

- Understand a business problem and translate it into a data problem
- Identify the target variable, and numerical vs. categorical features
- Investigate data quality: missing values, duplicates, invalid or suspicious values
- Make and justify data-cleaning decisions
- Perform and interpret univariate and bivariate EDA
- Analyze the target variable and relationships between features and the target
- Recognize possible data leakage
- Produce an ML-readiness table and a clean, documented dataset
- Push a well-organized notebook to GitHub

Module	Focus Area
1	The Business Problem — Customer Churn
2	Exploring the Dataset Structure
3	Investigating Data Quality
4	Data Cleaning Decisions
5	Univariate EDA
6	Target Variable Analysis

Module	Focus Area
7	Bivariate Analysis (Features vs. Churn)
8	Correlation & Feature Relationships
9	Data Leakage Awareness
10	ML-Readiness Checklist
11	Building the Clean Dataset
12	Key Findings & Final Reflection

Scenario & Dataset

Scenario

A telecommunications company is losing customers to competitors — this is called customer churn. The company wants to understand which customers are likely to leave, so that it can intervene early. Before any model can be built, the raw data must be understood, audited for quality issues, and explored. That is the entire job of this lab.

Dataset

You will work with the Telco Customer Churn dataset (Kaggle), containing approximately 7,043 customer records and 20+ columns covering demographics, subscribed services, contract & billing details, and a Churn label. The version provided contains simulated missing values, so the data-quality issues you find are genuine, not staged.

Category	Example Columns
Identifiers	customerID
Demographics	gender, SeniorCitizen, Partner, Dependents
Account / Contract	tenure, Contract, PaperlessBilling, PaymentMethod
Services	PhoneService, InternetService, OnlineSecurity, StreamingTV, ...
Billing	MonthlyCharges, TotalCharges
Target	Churn (Yes / No)

Before You Start

- Load the dataset and skim the column list — do not read ahead in this manual for answers.
- Create a new notebook: lab2_<yourname>_churn_eda.ipynb, with Markdown headings matching the parts below.
- You will push this notebook, and your cleaned dataset, to a GitHub repository at the end of the lab.

Part 1 — The Business Problem

Every ML project starts with a business question, not a dataset. Before writing any code, think about what the company actually wants to know, and what a single row of this dataset represents.

Tasks

1. In 2–3 sentences, state the business problem in your own words, and propose what the target variable likely is.

2. List 5 columns you expect to be useful predictors of churn, and 2 columns you expect to be useless as predictors — justify each choice in one line.
3. Write down 2 questions you personally want this data to answer about customers who churn.

Part 2 — Exploring the Dataset Structure

Before cleaning or analyzing anything, get an accurate picture of the dataset's shape and contents. Running a command is not the same as understanding its output — you must interpret what you see.

Concepts / Functions You May Find Useful

Concept / Function	Purpose
<code>df.shape</code> / <code>df.columns</code>	Row & column counts; column names
<code>df.dtypes</code> / <code>df.info()</code>	Data type of each column
<code>df.head()</code> / <code>df.sample(n)</code>	Inspect actual rows, including a random sample
<code>df[col].unique()</code> / <code>.nunique()</code>	Distinct values in a column
<code>df.describe(include='all')</code>	Summary statistics for numeric & categorical columns

Illustrative Example (generic — not the answer)

```
# Generic illustration only – apply the equivalent
# investigation to the churn dataset yourself.
df.shape
df.dtypes
df['some_column'].unique()
```

Tasks

4. How many customers and how many features are in this dataset? List the numerical features and the categorical features separately.
5. Identify any column(s) that are simply identifiers rather than predictive features, and any column(s) whose data type looks wrong for what it represents.
6. Pick 3 categorical columns and report their unique values — do any contain unexpected or inconsistent categories?

Part 3 — Investigating Data Quality

A dataset can look clean at a glance and still hide serious problems. You must actively search for missing values, duplicates, and suspicious values rather than assume there are none.

Concepts / Functions You May Find Useful

Concept / Function	Purpose
<code>df.isnull().sum()</code>	Count missing values per column
<code>(df.isnull().mean()*100)</code>	Percentage missing per column

Concept / Function	Purpose
<code>df.duplicated().sum()</code>	Count fully duplicated rows
<code>df['id_col'].duplicated().sum()</code>	Count duplicate IDs specifically
<code>df[col].value_counts()</code>	Frequency of each category / possible inconsistencies

Illustrative Example (generic — not the answer)

```
# Generic illustration only.
df.isnull().sum()
df.duplicated().sum()
```

Tasks

- Report missing values as both a count and a percentage per column. Which missing-value problems do you think require action, and which don't? Justify your reasoning.
- Check for duplicate rows and duplicate customer IDs separately — are they the same issue or different issues?
- Find at least one column with a suspicious, invalid, or inconsistently formatted value, and describe how you found it.

Part 4 — Data Cleaning Decisions

Cleaning is a series of justified decisions, not a fixed recipe. For every issue you fix, you must be able to explain why you chose that method over an alternative.

Concepts / Functions You May Find Useful

Concept / Function	Purpose
<code>pd.to_numeric(s, errors='coerce')</code>	Force a column to numeric, turning bad values into NaN
<code>df.fillna(value) / df.dropna()</code>	Fill or remove missing values
<code>df.drop_duplicates()</code>	Remove duplicate rows
<code>df[col].str.strip()</code>	Remove stray whitespace from text values
<code>df.astype(dtype)</code>	Convert a column to the correct data type

Tasks

- For every data-quality issue you found in Part 3, record: Problem → Decision → Method Used → Reason, in a small table in your notebook. Do not clean anything you cannot justify in writing.
- Apply your cleaning decisions to produce an intermediate cleaned dataframe, and confirm (with code output) that each issue is actually resolved.

Part 5 — Univariate EDA

Look at individual variables in isolation before looking at relationships. Every chart must be followed by a written interpretation — a graph with no explanation earns no credit.

Concepts / Functions You May Find Useful

Concept / Function	Purpose
<code>df[col].plot(kind='hist') / plt.hist()</code>	Distribution of a numerical column
<code>df.boxplot(column=col)</code>	Spread & outliers of a numerical column
<code>df[col].value_counts().plot(kind='bar')</code>	Frequency of a categorical column

Tasks

- Visualize the distribution of at least 2 numerical features and 2 categorical features. For each chart, write 1–2 sentences on what you learned.
- Do any numerical features show outliers or a skewed distribution? What might explain this?

Part 6 — Target Variable Analysis

The Churn column is what you will eventually predict. Before Lab 3, you need to understand its distribution — this has direct consequences for how a model should later be trained and evaluated.

Concepts / Functions You May Find Useful

Concept / Function	Purpose
<code>df['Churn'].value_counts()</code>	Count of each class
<code>df['Churn'].value_counts(normalize=True)</code>	Class proportions

Tasks

- Report the number and percentage of churned vs. non-churned customers. Is the target balanced or imbalanced?
- In 2–3 sentences, explain why class imbalance might matter when a classification model is trained later — do not describe how to fix it, just why it matters.

Part 7 — Bivariate Analysis: Features vs. Churn

Now connect individual features to the target. You choose the appropriate visualization or summary for each feature type — a categorical feature is not explored the same way as a numerical one.

Concepts / Functions You May Find Useful

Concept / Function	Purpose
<code>pd.crosstab(df[col], df['Churn'], normalize='index')</code>	Churn rate across categories
<code>df.groupby(col)['Churn']...</code>	Aggregate churn by group
<code>sns.boxplot(x='Churn', y=col, data=df)</code>	Compare a numerical feature across churn classes

Tasks

16. Investigate at least 2 categorical features (e.g. Contract, InternetService, PaymentMethod) against Churn. For each, record: Observation / Evidence / Possible Explanation.
17. Investigate at least 2 numerical features (e.g. tenure, MonthlyCharges) against Churn using the same Observation / Evidence / Possible Explanation format.

Part 8 — Correlation & Feature Relationships

Numerical features can be related to each other, not just to the target. Strongly correlated features can cause redundancy problems later, so it's worth knowing about them now.

Concepts / Functions You May Find Useful

Concept / Function	Purpose
<code>df.corr(numeric_only=True)</code>	Correlation matrix of numerical features
<code>sns.heatmap(corr_matrix, annot=True)</code>	Visualize the correlation matrix

Tasks

18. Compute and visualize the correlation matrix of the numerical features. Which pairs appear most correlated?
19. Does correlation imply causation here? Could any highly correlated features cause problems for a future model? Answer in 2–3 sentences.

Part 9 — Data Leakage Awareness

In Lab 3 you will train a classifier. Before that, you must check whether any column contains information that would not realistically be available at the moment of prediction — using such a column would make a model look artificially accurate.

Tasks

20. Inspect the columns for anything that looks like it could only be known after a customer has already churned, or that is purely an identifier. List any suspects and explain your reasoning.
21. In 2–3 sentences, explain what would go wrong if such a column were left in the model's input features.

Part 10 — ML-Readiness Checklist

Summarize every column's fate in a single table. This is the bridge between EDA and modeling — Lab 3 will use this table directly.

Template (fill in for every column in the dataset)

Column	Type	Role	Action	Reason
customerID	Identifier	—	Remove	...
tenure	Numerical	Feature	...	...

Column	Type	Role	Action	Reason
Contract	Categorical	Feature	...	...
Churn	Categorical	Target	Keep	...
...	...	...	...	...

Tasks

22. Complete the ML-readiness table for all columns in the dataset — not just the examples shown above.

Part 11 — Building the Clean Dataset

Bring together every cleaning decision into one final dataframe, `clean_df`, that Lab 3 can load directly. Do not split into train/test, encode categorical variables, or train any model — that happens next lab.

Tasks

23. Produce `clean_df` with correct data types, missing values handled, duplicates handled, and irrelevant identifier columns removed where justified. Save it (e.g. `clean_churn.csv`) alongside your notebook.
24. Print `df.info()` (or equivalent) for `clean_df` as final proof that every issue identified earlier has been resolved.

Part 12 — Key Findings & Final Reflection

Step back from the code and summarize what you actually learned about this dataset and this problem.

Five Important Findings — Template (repeat for each finding)

```
Finding 1:
Observation: ...
Evidence:    ...
Interpretation: ...
```

Final Reflection Questions

- What was the most important data-quality issue you found?
- What was the most interesting pattern you discovered?
- Which feature seems most useful for predicting churn?
- Which feature should probably not be used, and why?
- What preprocessing will be required before ML modeling (conceptually — do not implement it)?
- What would you do next if you had another 3 hours?

Tasks

25. Write your Five Key Findings using the Observation / Evidence / Interpretation format.
26. Answer all six Final Reflection questions in your notebook's concluding Markdown cell.

Bonus Tasks (Optional)

- Create one additional meaningful visualization not required above.
- Investigate a surprising customer subgroup you didn't expect to find.
- Calculate churn rate across a combination of 2+ customer segments at once.
- Identify one possible feature-engineering opportunity for Lab 3.
- Identify one possible source of bias in this dataset.

Submission Requirements

Submit the Following

- Jupyter/Colab notebook (.ipynb), organized with Markdown headings matching Parts 1–12.
- GitHub repository link with your notebook committed and pushed.
- Cleaned dataset (clean_churn.csv).
- Your Five Key Findings, clearly written.
- Your completed ML-readiness table.

Every major visualization must include a written interpretation. Notebooks containing only code, with no Markdown discussion, will not be accepted.

Grading Rubric

Criteria	Description
Data Understanding	Correctly identifies target, feature types, and dataset scope
Data Quality Investigation	Thorough, evidenced investigation of missing values, duplicates & invalid data
Data Cleaning Decisions	Every decision justified with a clear, sound reason
EDA Quality	Appropriate visualizations, correctly chosen for each feature type
Interpretation & Insights	Written interpretation accompanies every chart and finding
ML-Readiness & Documentation	Complete readiness table; clean, well-organized notebook

Penalties

- Copied Kaggle notebook or solution: significant mark deduction.
- Plots without written interpretation: no credit for that visualization.
- Unexplained or unjustified cleaning decisions: marks deducted in Part 4.
- Hard-coded / templated answers with no genuine analysis: marks deducted across affected sections.
- Training an ML model in this lab (not requested): marks deducted — save it for Lab 3.